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Quantitative analysis of cerebellar morphology in monotreme, metatherian and eutherian mammals

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ABSTRACT

To superficial inspection, the mammalian cerebellum appears to be a stereotypical structure that varies little in morphology across mammals. In the present study, the volumes of components of the corpus cerebelli, foliation of the cerebellar cortex and the volumes of the pontine and deep cerebellar nuclei have been measured and compared in three species of monotreme, 90 species of marsupial and 57 species of eutherian mammal. In all three mammalian groups, the volume of the corpus cerebelli scales isometrically with brain volume, and pontine nuclear volume also scales isometrically with cerebellar volume. The ratio of hemisphere to vermal cerebellar cortex is comparable in all mammals at small cerebellar volume, but elaboration of cerebellar hemispheres is largely confined to large cerebella of eutherian mammals. At small cerebellar volumes, diprotodontid metatherians have proportionally large cerebellar hemispheres compared to non-diprotodontid metatherians, and metatherian cerebella in general have a high volume of central white matter for a given cerebellar cortex volume compared to eutherians. The degree of foliation of the cerebellum scales similarly in therian mammals, but is relatively low in the monotremes for the volume of their corpus cerebelli. Among metatherians, cerebellar foliation is stronger among diprotodontid as compared to non-diprotodontids. Although the cerebellum has a similar structure in all mammals, there are subtle differences in structure between different mammal groups with possible functional implications.

1. Introduction

Monotremes (including the modern short- and long-beaked echidnas and the platypus) diverged from the therian lineage (leading to modern metatherian and eutherian mammals) as much as 220 million years ago (Yu et al., 2012). In our previous quantitative and magnetic resonance imaging studies of the brains of modern monotremes we found that they have a significantly lower volume of subiso- and periallocortical white matter in their forebrains compared to therians, suggesting that the forebrains of the modern monotremes have fewer association, commissural and/or projection connections than the forebrains of similarly sized therian mammals (Cherupalli et al., 2017; Ashwell and Gurovich, 2019). This could suggest substantially different information processing in the forebrains of monotremes, compared to therians.

Apart from the forebrain cortical systems, the other laminated brain structure engaging in looped and reciprocal connections with diverse parts of the brain is the cerebellum and its precerebellar afferent nuclei (cerebellar/precerebellar system) (Butler and Hodos, 2005). Although the cerebellum and its precerebellar input nuclei were traditionally seen (primarily on the basis of clinical studies) as a motor learning and regulatory system, it is currently recognised that the cerebellum has a much more diverse role. The lateral cerebellum (cerebellar hemispheres and dentate nucleus) may regulate cognitive information processing by

Abbreviations: 1Cb to 10Cb, lobules of cerebellar vermis; 4V, fourth ventricle; cbw, deep cerebellar white matter; CIC, central nucleus of inferior colliculus; Crus1, crus 1 of ansiform lobule; Crus2, crus 2 of ansiform lobule; DCIC, dorsal cortex of inferior colliculus; DLL, dorsal nucleus of lateral lemniscus; DMTg, dorsomedial tegmental nucleus; DTg, dorsal tegmental nucleus; ECIC, external cortex of inferior colliculus; ESC, external surface of cerebellum; Fl, flocculus; GrCb, granule cell layer of cerebellar cortex; ILL, intermediate nucleus of lateral lemniscus; LDTg, laterodorsal tegmental nucleus; LDTgV, laterodorsal tegmental nucleus, ventral; lfp, longitudinal fibres of the pons; ll, lateral lemniscus; LPB, lateral parabrachial nucleus; m5, motor root of trigeminal nerve; mcp, middle cerebellar peduncle; mlf, medial longitudinal fasciculus; MoCb, molecular layer of cerebellar cortex; MPB, medial parabrachial nucleus; PFlD, paraflocculus dorsal; PFlV, paraflocculus ventral; pfs, parafloccular sulcus of cerebellum; plf, posterolateral cerebellar fissure; Pn, basilar pontine nuclei; PnO, pontine oral reticular nucleus; prf, primary cerebellar fissure; psf, posterior superior fissure of the cerebellum; PSC, pial surface of cerebellum; RtTg, reticulotegmental nucleus of pons; s5, sensory root of trigeminal nerve; Sag, sagulum nucleus; scp, superior cerebellar peduncle; Sim, simplex lobule of cerebellum; Simf, simplex fissure of cerebellum; VLL, ventral nucleus of lateral lemniscus

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automating sequences of thoughts and actions (Schmahmann, 1991; Smaers et al., 2018), thereby supporting many aspects of intelligent behaviour, including working memory, executive function and the development of models for behavioural learning (Strick et al., 2009; Smaers et al., 2018). In other words, the cerebellum may play an important part in those complex forebrain-mediated behaviours that have been elaborated during mammalian evolution as a result of cerebral cortex expansion. This notion is consistent with the concerted expansion of the cerebellum in step with the cerebral cortex (Herculano-Houzel, 2010).

Although the mature cerebellum and its afferent precerebellar nuclei appear superficially similar across all mammals (Butler and Hodos, 2005; Ashwell et al., 2007), there are some significant developmental differences between monotremes and therians. In a previous study, it was found that the structure of the developing monotreme cerebellum and the arrangement of transitory neuronal populations are similar to those reported for fetal and neonatal eutherian mammals, but the time-course of the key events of later cerebellar development (e.g. granule cell migration, primary fissuration and foliation) is spread over a much longer period in monotremes (Ashwell, 2012; Ashwell et al., 2019).

Our previous studies also showed that there are significant differences within therians in the timing of cerebellar and precerebellar development (Ashwell et al., 2019). We found that, with respect to absolute body size, developing metatherians in general are in advance of eutherian mammals for attainment of some early cerebellar and precerebellar milestones, although this was not the same for all metatherians and was not sufficient to provide a fully mature locomotor system at birth. In particular, dasyurids appear to be significantly more advanced than other metatherians in cerebellar development. This is in spite of young dasyurids having low internal aortic luminal area, and narrower cerebral feeder arteries compared to young non-dasyurid metatherians (Ashwell and Shulruf, 2016), which would imply poor transfer of nutrients to the developing brain. Other authors have also noted that metatherian cerebella have more neurons than non-primate eutherian cerebella of a similar mass (Dos Santos et al., 2017), based on a sample of 10 metatherian species.

Previous comparative studies of cerebellar structure have focussed on eutherian mammals (Jacobs et al., 2014; Smaers et al., 2018), mainly because only limited sectioned metatherian brain material has been available (e.g. of didelphids and occasional macropods) and monotremes are often excluded altogether. This means that metatherians and monotremes are often under-represented in such analyses. In the present study it was possible to use the resources of the Nelson metatherian and monotreme brain collection (held at the Australian National Wildlife Collection, CSIRO Canberra) to analyse cerebellar morphology across all three subclasses of extant mammals. The aims of the study were to determine whether all mammals share similar scaling of cerebellar morphological features (e.g. cerebellar cortical volume, cerebellar white matter volume, deep cerebellar nuclear volume, pontine/reticulotegmental nuclear volume, hemispheric enlargement and cerebellar foliation; see Table 1) as would be expected if the cerebellar/precerebellar system were functioning similarly across all mammals, and to try to relate any differences to evolutionary history and behaviour. Table 1 sets out the cerebellar morphological variables measured in the various species and the rationale for inclusion of each variable.

The analysis has been focused on allometric relationships *within* the brain (i.e. scaling of cerebellum with whole brain, or scaling of cerebellar cortex or deep nuclei with whole cerebellum) rather than with whole body size, because analysis of scaling against body size can be confounded by variation in abdominal visceral mass associated with diet.

2. Materials and Methods

This study is mainly based on sectioned metatherian and monotreme brains from the Nelson Brain Collection held at the CSIRO Australian National Wildlife Collection in Canberra (please see Ashwell and Gurovich, 2019 for details) and the author's collection. These brains include 3 species of monotreme (two tachyglossids and one ornithorhynchid) and 90 species of metatherians (eight didelphimorphids, one microbiotheriid, 18 dasyuromorphids, five peramelemorphids, and 58 diprotodontids) (see Table 2 and Supplementary Table 1). Two specimens of monotreme were from published atlases (Ashwell and Hardman, 2013a,b) and one specimen of diprotodontid (*Macropus eugenii*) was also from a published atlas (Ashwell and Marotte, 2010).

The brains of the eutherian mammals studied here were from the Neuroscience Library collection (scaled frontal plane images available at NeuroscienceLibrary.org) and were chosen to represent a broad range of mammalian superorders or orders (Afrotheria, Carnivora, Cetartiodactyla, Chiroptera, Dermoptera, Lagomorpha, Lipotyphla, Perissodactyla, Primata, Rodentia, Scandentia, and Xenarthra; see Table 2). Those brains have been embedded in celloidin and sectioned at thicknesses of 25 to 40 μm. All sections used for this analysis were Nissl-stained. The website provides calibrated images of the whole fixed, but undehydrated, brains, which allow dimensions of the cerebellum to be measured. These could then be applied to dehydrated section images to calibrate photomicrographs and calculate shrinkage (Ashwell and Gurovich, 2019). One specimen was from a published brain atlas (*Callithrix jacchus* – Hardman and Ashwell, 2012). The analysis used only specimens where sections from the entire rostrocaudal extent of the cerebellum were available. For *Dromiciops gliroides* and some diprotodontid metatherian specimens (*Acrobates pygmaeus*, *Burramys parvus*, *Cercartetus nanus*, *Gymnobelideus leadbeateri*, *Hemibelideus lemuroides*, *Macropus eugenii*, *Onychogalea fraenata*, *Petaurus australis, Petrogale brachyotis*, *Phascolarctos cinereus*, *Pseudocheirus cinereus*, *Tarsipes rostratus*, *Vombatus ursinus*), sections were photographed with an *Olympus BHS2* microscope with an SPlan objective (magnification of 4x) and an FK photo projection lens (power of 5x). The imaging sensor was a *QImaging MicroPublisher* 3.3 million pixel, real-time viewing camera. Calibration was achieved by scanning a standard 1.0 mm graduated slide scale (10 μm increments) in two orthogonal directions. The images were stitched with Adobe Photoshop CS5.1 to create montages of frontal sections through the brain at intervals of 300 μm from the rostral tip of the olfactory bulb to the cervical spinal cord. Calibration of photomontaged images was further checked after stitching by comparing the actual section dimensions with expected dimensions from calibrated component images. For all other metatherian and monotreme specimens, slides from the Nelson Collection were scanned at 1,200 dpi on an *Epson Perfection V370 Photo* flatbed scanner (white LED light source, matrix CCD, 12 line with on-chip microlenses). Calibration was achieved by scanning a 297 mm long, 1.0 mm graduated scale in two orthogonal directions.

Measurements were made on one side of the brain only (right hemisphere as seen on the slide) using *Fiji* (ImageJ2, Rasband, 2014) and multiplied by two for whole brain values. Intervals between sections were used to calculate volumes and areas across the rostrocaudal extent of the forebrain by Cavalieri's basic estimator (Rosen and Harry, 1990). Intervals between the sections that were measured ranged from 0.20 mm for the smallest brains, up to 2.50 mm for the largest brains, to provide 10 to 20 sections per specimen for analysis.

Foliation index was calculated as the ratio of total pial surface cerebellar area (PSC) of the corpus cerebelli (i.e. the body of the cerebellum) divided by the external surface cerebellar area (ESC) (see Table 1). The PSC includes all sulci and fissures of cerebellar cortex, whether facing the subarachnoid space or the fourth ventricle. The ESC is the surface of a multi-sided convex polygon enclosing the external surface of cerebellum in contact with either the subarachnoid space or the fourth ventricle lumen.

All measurements of brain sections were converted to those for fixed, but undehydrated brains. For celloidin or paraffin-embedded brains, linear and volume shrinkage was calculated by comparing pre-

Table 1
Cerebellar volumetric and areal variables measured in the present study.

Variable	Definition	Rationale/purpose
Cerebellar vermal volume (lobules 1 to 9)	Volume of the cerebellar cortex (including molecular MoCb, Purkinje and granular cell GrCb layers) in the cerebellar vermis, lobules 1 to 9 only. This region includes the spinocerebellum functional area responsible for processing spinal cord input to co-ordinate proximal musculature.	To assess cortical volume of the cerebellar vermis (lobules 1 to 9) and for calculation of total cerebellar cortical volume.
Cerebellar hemisphere volume (excluding flocculus)	Volume of cerebellar cortex (including molecular, Purkinje and granular cell layers) in the hemispheres. Note that it includes paraflocculus, but excludes flocculus. The simplex fissure and medial border of crus 1 and 2 of the ansiform lobule provided medial borders of the hemisphere. This region includes cerebropontocerebellum responsible for co-ordination of distal musculature.	To assess the degree of hemispheric specialisation (hemisphere and paraflocculus) relative to vermal cortex and for calculation of total cerebellar cortical volume. Hemispheric expansion is indicative of enlargement of the cerebropontocerebellum, suggesting enhanced function of fine distal musculature.
Flocculonodular lobe volume	Volume of cerebellar cortex in flocculus and nodule (including molecular, Purkinje and granular cell layers) responsible for processing of vestibular input to co-ordinate eyes and head movement.	To assess size and functional significance of vestibulocerebellar cortex (nodulus or vermal lobule 10 and flocculus) and for calculation of total cerebellar cortical volume.
Cerebellar deep white matter volume	Volume of the deep white matter (cbw) within the corpus cerebelli (excluding deep cerebellar nuclei)	For assessment of connections to and from cerebellar neuronal populations and for calculation of total cerebellar cortical volume.
Volume of deep cerebellar nuclei	Volume of deep cerebellar nuclei, including medial, intermediate and lateral nuclei, and interstitial cell group (i.e. excluding nucleus Y and vestibulocerebellar nucleus which are members of the vestibular nuclear complex)	For assessment of size of nuclei responsible for non-vestibular output from cerebellum and for calculation of total cerebellar cortical volume.
External surface of cerebellum (ESC)	Surface of multi-side convex polygon enclosing external surface of cerebellum in contact with subarachnoid space and fourth ventricle lumen	For calculation of foliation index (PSC/ESC).
Pial surface of cerebellum (PSC)	Extent of pial surface, including all sulci and fissures of cerebellar cortex, whether facing subarachnoid space or fourth ventricle	For calculation of foliation index (PSC/ESC).
Volume of pontine nuclei	Volume of pontine and reticulotegmental nuclei, excluding longitudinal and transverse fibres of the pons	To assess role of cerebropontocerebellar circuitry.

and post-dehydration brain dimensions as measured with *Fiji* from photographs provided at NeuroscienceLibrary.org, or stated in Nelson Collection records. Mean linear shrinkage (± SD) was found to be 0.738 (± 0.052) for 47 species, equivalent to a volume shrinkage factor of 0.403. There was no significant correlation between initial brain size and the magnitude of linear shrinkage ($r = -0.111$, $p = 0.228$). Brain volume was either taken from stated values in atlases or collection records, or calculated from the measurements of serial sections through the entire brain, with conversion for shrinkage.

Least squares linear regression of cerebellar and precerebellar variables (Table 1) against brain volume and/or other measures, along with calculation of 95% confidence intervals was performed using *Microsoft Excel Mac2011 v14.0.0*. Plots were prepared using *Adobe Illustrator CS5.1*. Analysis of the scaling of area against volume was performed by first converting both variables to single linear dimensions (i.e. the square root of area was plotted against the cube root of volume). Logarithmic scales were used for analysis of scaling of volumes because of the large range in values.

3. Results

Fig. 1 illustrates the boundaries and method of measurement for transverse sections through the cerebellum of a representative diprotodontid metatherian (*Macropus eugenii*). For the purposes of the present study "pontine nuclei" was taken to include both the basilar pontine and reticulotegmental pontine nuclei, since both engage in cerebro(ponto)cerebellar looped connections (Cicirata et al., 2005). The reticulotegmental nucleus receives cortical afferents from motor or sensorimotor cortical areas and the basilar pontine nuclei receive afferents principally from the sensory and associative cortical areas (Cicirata et al., 2005).

3.1. Quantitative comparison of cerebellar cortical morphology among monotreme, metatherian and eutherian mammals

In metatherian and eutherian mammals, the volume of the corpus cerebelli scaled isometrically with brain volume, i.e. a doubling of brain volume was accompanied by a doubling of cerebellar volume (Fig. 2; gradient for eutherians and metatherians was 0.932 and 1.018, respectively). This is similar to that observed by Herculano-Houzel (2010) with slight differences in slope that would be within expected biological variation. Herculano-Houzel has shown that there is co-ordinated scaling of cerebellar neurons with cortical neurons such that there are on average 3.6 cerebellar neurons for every cortical neuron.

There was no apparent difference between diprotodontid and non-diprotodontid metatherians with respect to cerebellar volume in that cerebellar volume scaled almost isometrically with brain volume in both groups (not illustrated; gradients of 0.974 and 1.073, respectively). Cerebellar volume also scaled isometrically among monotremes (gradient of 1.008), but the absolute values for cerebellar volume among monotremes are approximately 60% of therians for a given brain volume. At this stage it is not possible to say if monotremes have fewer cerebellar neurons than eutherians of the same cerebral size.

The volume of the flocculonodular lobe cerebellar cortex (flocculus and 10Cb together) scaled close to isometrically among therians (Fig. 3), but for a given cerebellar volume the size of the flocculonodular lobe was 20 to 25% higher among metatherians than eutherians, suggesting a greater devotion of cerebellar cortical tissue to vestibulocerebellar functions among metatherians. Flocculonodular cortex volume was low in the long-beaked echidna (*Zaglossus bruijnii*) for its cerebellar size.

The volume of spino- and cerebro(ponto)cerebellar cortex (i.e. cerebellar cortex not part of the flocculus or nodulus) was plotted against cerebellar volume (Fig. 4) for all mammalian groups. There was a closely isometric relationship between spino- and pontocerebellar cortex volume and cerebellar volume among all three groups of mammals (gradients of 1.021, 0.972 and 0.992 for monotremes, metatherians and eutherians, respectively) across a wide range of cerebellar volume. The spino- and cerebro(ponto)cerebellar cortex consistently take up 75 to 80% of the corpus cerebelli across a range of cerebellar volumes from 10 to 100,000 mm^3 in all three mammalian groups.

Enlargement of the cerebellar hemispheres is seen among eutherian species with large corpus cerebelli (Smaers et al., 2018), but this has never been analysed quantitatively among metatherians and

Table 2
List of species used in the present study.

Mammalian group		Species	No. specimens (N)
Monotremata	Ornithorhynchidae	*Ornithorhynchus anatinus*	3
	Tachyglossidae	*Tachyglossus aculeatus*	3
		Zaglossus bruijnii	1
Metatheria	Didelphimorphia	*Caluromys derbianus*	1
		Didelphis virginiana	1
		Didelphis marsupialis	1
		Didelphis albiventris	1
		Marmosa impavida	1
		Monodelphis brevicaudata	1
		Rhyncholestes raphanurus	1
		Thylamys elegans	1
	Microbiotheriidae	*Dromiciops gliroides*	1
	Dasyurimorphia	*Antechinomys laniger*	1
		Antechinus flavipes	1
		Antechinus minimus	1
		Antechinus swainsonii	1
		Dasycercus blythi	1
		Dasykaluta rosamundae	1
		Dasyuroides byrnei	1
		Dasyurus geoffroyi	1
		Dasyurus hallucatus	1
		Dasyurus maculatus	1
		Dasyurus viverrinus	1
		Phascogale tapoatafa	1
		Planigale maculata	1
		Pseudantechinus bilarni	1
		Sarcophilus harissii	1
		Sminthopsis crassicaudata	1
		Sminthopsis leucopus	1
		Sminthopsis murina	1
	Peramelemorphia	*Echymipera kalubu*	1
		Isoodon macrourus	1
		Isoodon obesulus	1
		Perameles gunnii	1
		Perameles nasuta	1
	Diprotodontia	*Acrobates pygmaeus*	1
		Aepyprymnus rufescens	1
		Bettongia gaimardi	1
		Burramys parvus	1
		Cercarteus concinnus	1
		Cercartetus lepidus	1
		Cercartetus nanus	1
		Dendrolagus goodfellowi	1
		Dendrolagus lumholtzi	1
		Distoechurus pennatus	1
		Dorcopsis luctuosa	1
		Gymnobelideus leadbeateri	1
		Hemibelideus lemuroides	1
		Hypsiprymnodon moschatus	1
		Lagorchestes conspicillatus	1
		Lasiorhinus latifrons	1
		Macropus agilis	1
		Macropus antilopinus	1
		Macropus bernardus	1
		Macropus dorsalis	1
		Macropus eugenii	1
		Macropus irma	1
		Macropus parma	1
		Macropus parryi	1
		Macropus rufus	1
		Onychogalea fraenata	1
		Onychogalea unguifera	1
		Petaurus australis	1
		Petaurus norfolcensis	1
		Petrogale brachyotis	1
		Petrogale concinna	1
		Petrogale godmani	1
		Petrogale herberti	1
		Petrogale inornata	1
		Petrogale lateralis	1
		Petrogale penicillata	1
		Petrogale puella (assimilis)	1
		Petrogale purpureicollis	1
		Petropseudes dahli	1
		Phalanger gymnotis	1
		Phalanger maculatus	1
		Phalanger vestitus	1
		Phascolarctos cinereus	1
		Potorous tridactylus	1
		Pseudocheirus canescens	1
		Pseudocheirus cinereus	1
		Pseudocheirus forbesi	1
		Pseudocheirus peregrinus	1
		Pseudochirops archeri	1
		Pseudochirops cupreus	1
		Pseudochirulus herbertensis	1
		Tarsipes rostratus	1
		Thylogale stigmatica	1
		Thylogale thetis	1
		Trichosurus arnhemensis	1
		Trichosurus caninus	1
		Wallabia bicolor	1
		Vombatus ursinus	1
Eutheria	Afrosoricida	*Tenrec ecaudatus*	1
	Carnivora	*Ailurus fulgens*	1
		Bassariscus astutus	1
		Canis familiaris (Basenji)	1
		Crocuta crocuta	1
		Cynictis penicillata	1
		Felis cattus	1
		Galictis vittata	1
		Mustela erminea	1
		Mustela putorius	1
		Nasua narica	1
		Neovison vison	1
		Panthera leo	1
		Phoca vitulina	1
		Taxidea taxus	1
		Ursus maritimus	1
		Zalophus californianus	1
	Cetartiodactyla	*Bos taurus indicus*	1
		Capra hircus domestica	1
		Equus quagga burchellii	1
		Lama glama	1
		Odocoileus virginianus	1
		Pecaru tacaju	1
		Sus scrofa domesticus	1
		Tursiops truncatus	1
	Cingulata	*Dasypus novemcinctus*	1
	Chiroptera	*Pteropus giganteus*	1
	Dermoptera	*Cynocephalus volans*	1
	Eulipotyphla	*Sorex araneus*	1
		Erinaceus europaeus	1
		Scalopus aquaticus	1
	Hyracoidea	*Procavia capensis*	1
	Lagomorpha	*Oryctolagus cuniculus*	1
	Macroscelidea	*Elephantulus myurus*	1
	Pilosa	*Choloepus didactylus*	1
		Myrmecophagia tridactyla	1
		Tamandua tetradactyla	1
	Primata	*Alouatta palliata*	1
		Aotus trivirgata	1
		Callicebus moloch	1
		Callithrix jacchus	1
		Cebus capucinus	1
		Eulemur mongoz	1
		Mandrillus sphinx	1
		Pan troglodytes	1
		Perodicticus potto	1
		Saimiri sciureus	1
		Semnopithecus entellus	1
	Rodentia	*Aplodontia rufa*	1
		Castor canadensis	1
		Cavia porcellus	1
		Chincella lanigera	1
		Erithizon dorsatum	1
		Sciurus carolinensis	1

(*continued on next page*)

Table 2 (*continued*)

Mammalian group	Species	No. specimens (N)
	Thomomys talpoides	1
	Aplodontia rufa	1
Scandentia	*Tupaia glis*	1
Sirenia	*Trichechus manatus latirostris*	1

monotremes. The ratio of the volume of cerebellar cortex in the hemisphere to the vermis was plotted against cerebellar volume for monotreme, metatherian and eutherian mammals and is shown in Fig. 5a. Cerebellar hemispheric enlargement was similar among both eutherians and metatherians up to a cerebellar volume of 1,000 mm^3. Some eutherians (e.g. marine mammals such as *Phoca vitulina*, *Tursiops truncatus* and *Zalophus californianus* as well as the large-brain primate *Pan troglodytes*) showed pronounced enlargement of the cerebellar hemispheres beyond that expected for their cerebellar volume, as has been reported previously (Smaers et al., 2018). The reasons for this are not clear, but may be associated with maximising cerebellar neuron number to maintain functionality of an enlarged neocortex/cerebellum ensemble in these larger-brained species (Herculano-Houzel, 2010). By contrast, the proportion of cerebellar cortex in the hemispheres among metatherians did not rise above 66%, even in the largest metatherian cerebella. Nevertheless, there was a difference between diprotodontid and non-diprotodontid metatherians in the degree of hemispheric enlargement with increasing cerebellar volume (Fig. 5b). The proportion of cerebellar cortex in the hemispheric did not vary with cerebellar volume among diprotodontia, staying at about 40 to 65% across a wide range of cerebellar volumes (20 to 7,000 mm^3), but increased from 20 to 50% with increasing cerebellar volume among non-diprotodontids (please see regression lines, confidence intervals and results of regression analysis in Fig. 5b). Hemispheric specialisation was very poorly

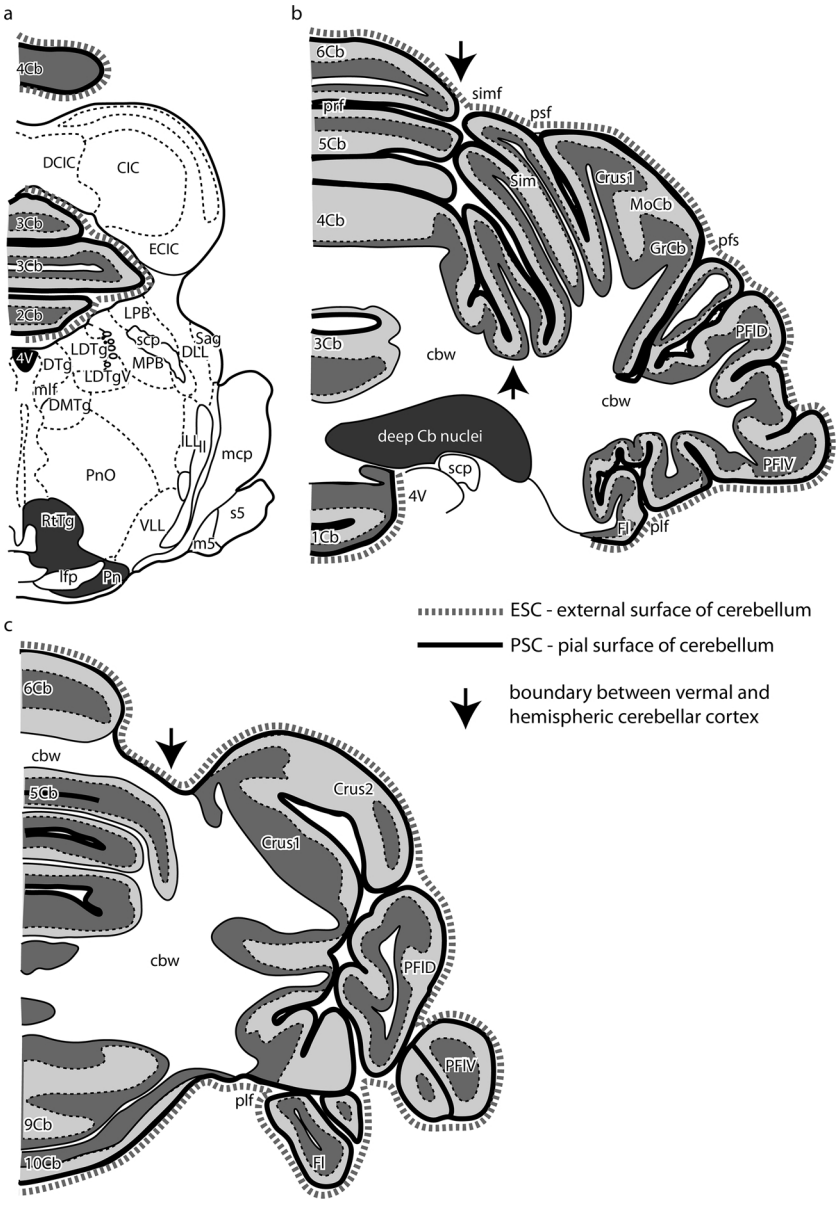


Fig. 1. Line diagrams of a series of frontal sections through the cerebellum of a representative diprotodontid metatherian, the tammar wallaby (*Macropus eugenii*) (a – rostral vermis and pontine/reticulotegmental nuclei; b – level of deep cerebellar nuclei; c – caudal vermis and hemispheres), showing how areas and volumes were calculated. Please refer to Table 1 for definitions of measurements. For the more caudal levels (b, c), the vertically oriented arrow marks the boundary between vermal and hemispheric cortex (for purposes of this study, simf and medial border of crus 1 and 2 of ansiform lobule), as used here.

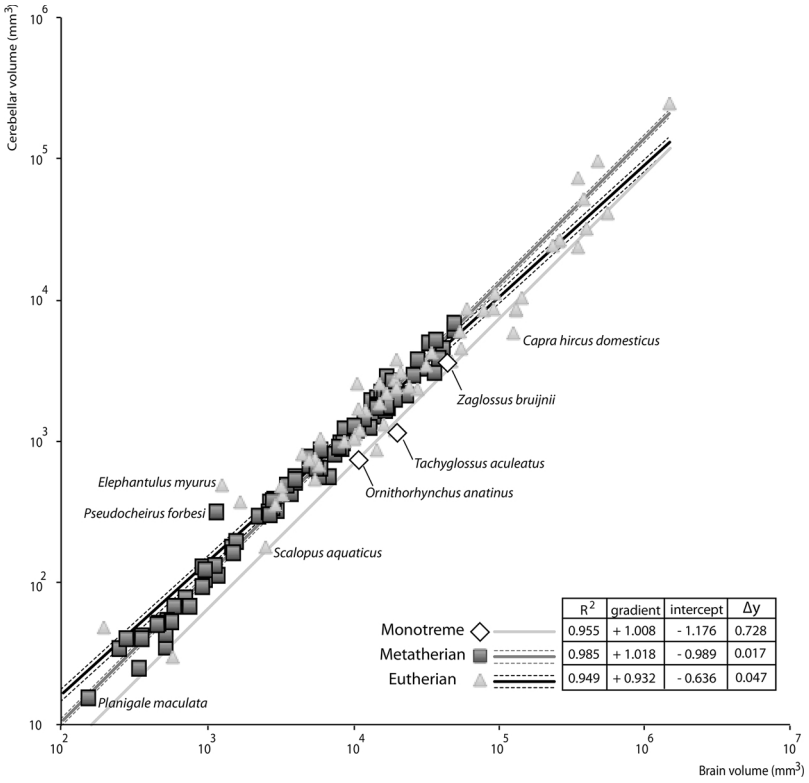


Fig. 2. Plot of total cerebellar volume in mm^3 against brain volume in mm^3 for monotremes, metatherians and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. For all mammals, the scaling relationship is close to isometric, in that cerebellar volume increases in proportion to brain size increase. Monotremes have a slightly smaller corpus cerebelli for their brain volume compared to most therians. The smallest cerebellum in the sample was that for the dasyurid *Planigale maculata*. Some eutherian species have small corpus cerebelli for brain size (e.g. *Scalopus aquaticus* and *Capra hircus domesticus*), whereas others have large corpus cerebelli (e.g. *Elephantulus myurus*). Note that the intercepts in the table are $\log_{10}$ of the cerebellar volume in mm^3 for a brain volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regression) is also $\log_{10}$ of cerebellar volume in mm^3.

developed in small non-diprotodontid carnivores with small corpus cerebelli (e.g. *Antechinus minimus*, *Antechinomys laniger* and *Sminthopsis murina*; Fig. 5b).

The degree of folding of cerebellar cortex (foliation index or fi; ratio of PSC/ESC; see Methods and Table 1) appears to increase with cerebellar size among mammals and birds in general and may have sensorimotor functional significance (Welker, 1990; Iwaniuk et al., 2006), but this has never been analysed quantitatively among a large sample of

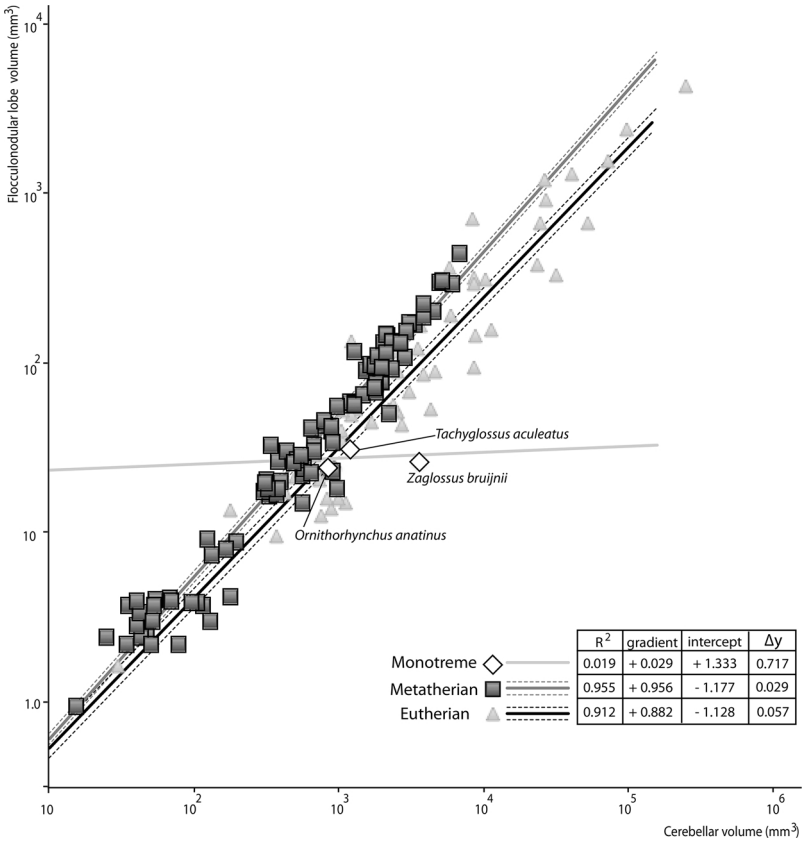


Fig. 3. Plot of flocculonodular lobe volume (flocculus plus 10Cb) in mm^3 against corpus cerebelli volume in mm^3 for monotremes, metatherians and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. For metatherians, scaling was close to isometric (gradient of + 0.956), but scaling was below isometric for eutherians and monotremes. Flocculonodular lobe volume was slightly higher among metatherians than eutherians and particularly low in the long-beaked echidna (*Zaglossus bruijnii*) for its corpus cerebelli volume. Note that the intercepts in the table are $\log_{10}$ of the flocculonodular lobe volume in mm^3 for a corpus cerebelli volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regression) is also $\log_{10}$ of flocculonodular volume in mm^3.

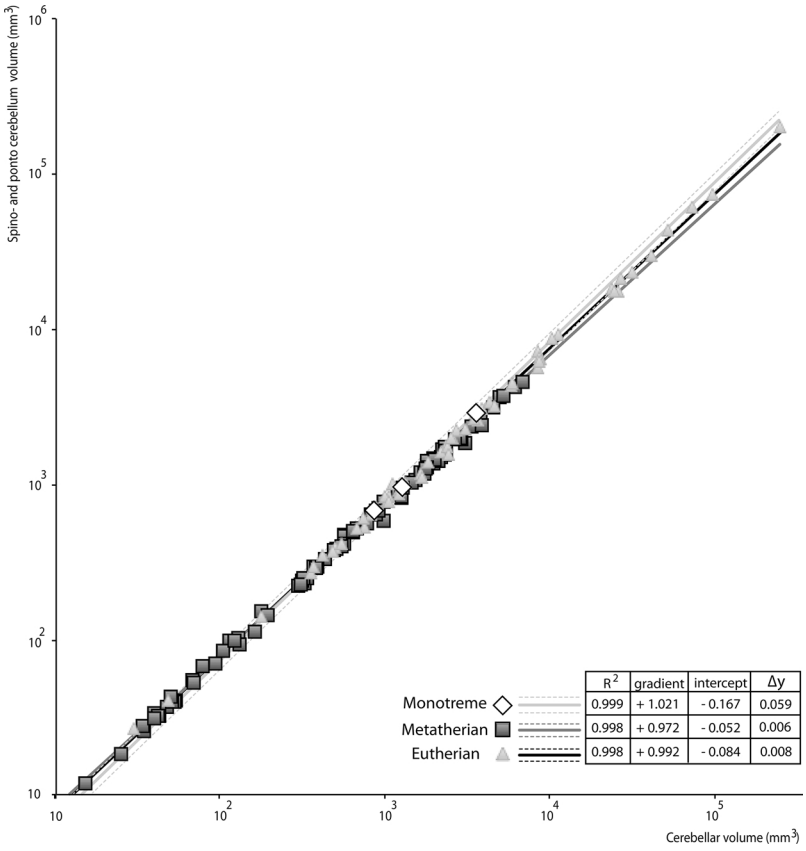


Fig. 4. Plot of spino- and cerebro(ponto)cerebellar cortex volume (1Cb to 9Cb inclusive plus cerebellar hemispheres, including paraflocculus) in mm^3 against corpus cerebelli volume in mm^3 for monotremes, metatherians and eutherians. The 95% confidence interval for the linear regression is shown as dashed lines for the monotreme data. The confidence intervals for eutherian and metatherian data are too close to the respective regression lines to be visible. Note that the scaling relationship is very close to isometric for all three groups of mammals. The intercepts in the table are log_{10} of the spino- and pontocerebellar cortex volume in mm^3 for a corpus cerebelli volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regression) is also log_{10} of spino- and pontocerebellar cortex volume in mm^3.

metatherians and monotremes. The degree of foliation of the cerebellar cortex was plotted against total cerebellar cortex volume for monotreme, metatherians and eutherians (Fig. 6a). Among therians in general, foliation rose with cerebellar cortical volume, with some outliers with high (*Ailurus fulgens*, *Capra hircus domesticus*, *Scalopus aquaticus*) or low foliation (*Bos taurus indicus)* for their cerebellar cortical volume. By contrast, among the three species of monotreme, foliation was low for the platypus and (for monotremes) did not rise across a 7 or 8-fold range of cerebellar size for the adult individuals of the three species, such that the cerebellar cortex of the adult long-beaked echidna was even less foliated (fi of 1.95) than that of the adult platypus (fi of 2.10). Among metatherians (Fig. 6b), a difference was apparent between diprotodontid and non-diprotodontid species. Although foliation increased with cerebellar cortical volume in both, foliation was slightly higher among diprotodontid compared to non-diprotodontid species at both low and high cerebellar cortical volume (please see regression lines for diprotodontids versus non-diprotodontids, confidence intervals and results of regression analysis in Fig. 6b). The least-foliated cerebellum was that for the dasyurid *Dasykaluta rosamundae*.

The square root of the pial surface area of the cerebellum was plotted against the cube root of brain volume for all three mammalian groups (Fig. 7) as a indicator of cerebellar cortical processing for a given brain size. Consistent with the low foliation and small corpus cerebelli of the monotremes, cerebellar pial surface area is approximately 40% lower among the monotremes for a given brain size, suggesting a possibly lower processing capacity for the cerebellar cortex of monotremes.

3.2. Comparison of scaling of cerebellar white matter volume with cerebellar cortical volume among mammals

Among therians, cerebral white matter is known to scale with iso- and periallocortical grey matter volume with a gradient of 1.073, but monotremes have a significantly lower cerebral white matter volume than therians of the same cerebral grey matter volume (Ashwell and Gurovich, 2019). In order to determine if cerebellar white matter volume scales differently among therians and prototherians, cerebellar white matter volume was plotted against cerebellar cortical volume among monotreme, metatherians and eutherians (Fig. 8). Among both metatherians and eutherians, cerebellar white matter scales with cerebellar cortical volume at above isometric (positive allometry; metatherian – 1.197; eutherian – 1.134), but cerebellar white matter volume is 68% higher among metatherians compared to eutherians at a cerebellar cortical volume of 10,000 mm^3. Among monotremes, cerebellar white matter scales close to isometrically with cerebellar cortical volume (gradient of 1.024), but is similar in volume to that in eutherians at the cerebellar cortical volumes seen among the extant monotremes.

3.3. Scaling of pontine and deep cerebellar nuclear volume with cerebellar cortical volume among mammals

The pontine nuclei are a key part of the looped cerebro(ponto)cerebellar circuit mediating planning and control of fine movement, particularly of the extremities, and may also play an important role in cognition (Strick et al., 2009; Smaers et al., 2018). The volume of the pontine nuclei has been plotted against cerebellar cortical volume in Fig. 9 and the relationship is close to isometric for all three mammalian groups (gradients of 1.084, 1.005 and 1.038 for monotremes, metatherians and eutherians, respectively). There were no apparent differences either between monotremes and therians, or between metatherians and eutherians, in this relationship across a 100,000-fold range in cerebellar cortical volume.

The deep cerebellar nuclei are the sole output neurons for the ponto- and spinocerebellar components of the cerebellum. The volume of the deep cerebellar nuclei scaled isometrically with cerebellar cortical volume among all three groups of mammals (gradients of 1.015, 0.978 and 0.952 for monotremes, metatherians and eutherians, respectively;

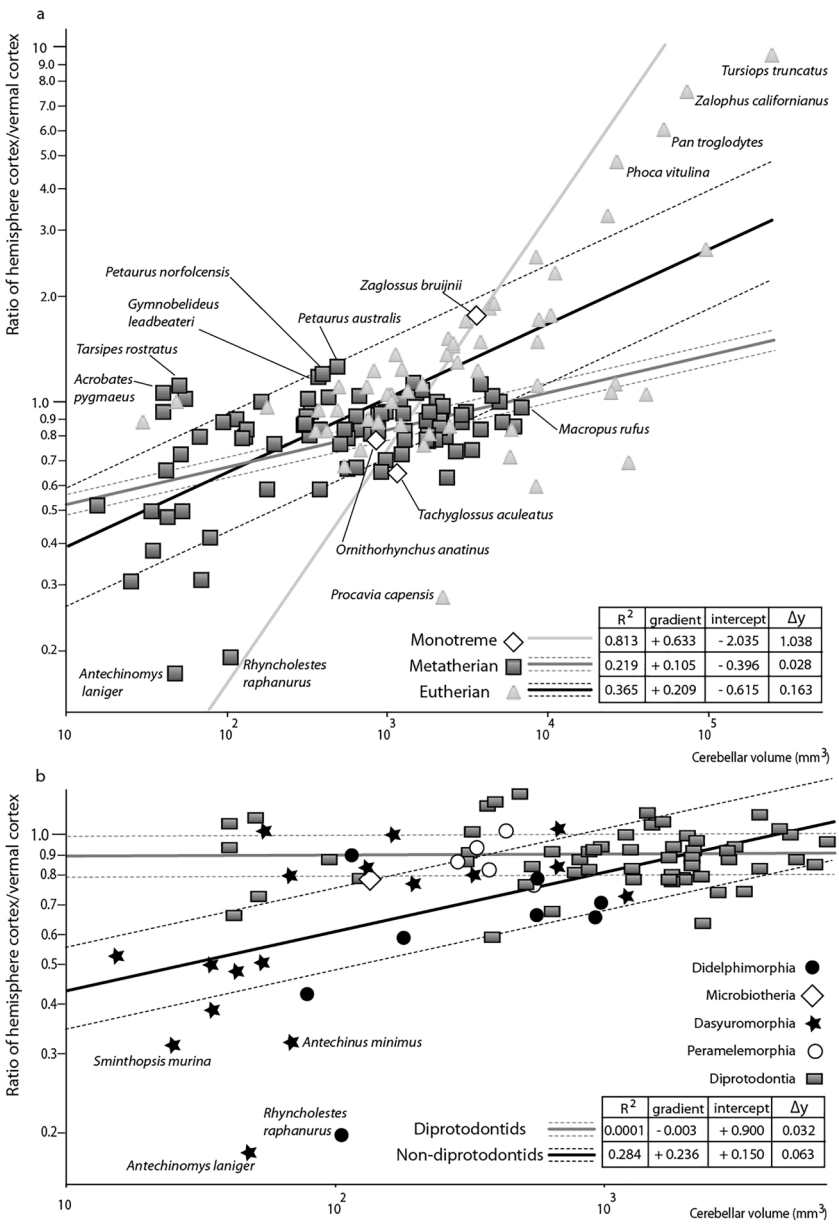


Fig. 5. Plots of the ratio of hemisphere cortex/vermal cortex (excluding 10Cb) against corpus cerebelli volume in mm^3 for monotremes, metatherians and eutherians (a) and for metatherian subgroups (b). The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Several eutherian species with large corpus cerebelli (in particular marine mammals such as *Phoca vitulina*, *Tursiops truncatus*, *Zalophus californianus*, as well as large-brain primates such as *Pan troglodytes*) also have high hemisphere/vermis ratios. Metatherians such as *Rhyncholestes raphanurus* and *Antechinomys laniger* as well as the eutherian *Procavia capensis* have very small cerebellar hemispheres. Among diprotodontia (b), the hemisphere/vermal ratio is relatively unchanged across a 200-fold range in cerebellar volume; and dasyuromorphia and didelphimorphia have relatively small cerebellar hemispheres for their cerebellar volume, compared to diprotodontia. Note that the intercepts in the table in a) are $\log_{10}$ of the hemisphere/vermal ratio for a corpus cerebelli volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regression) is also $\log_{10}$ of the hemisphere/vermal ratio.

Fig. 10a). On the other hand, in the middle range of cerebellar size (cerebellar cortical volumes of 200 to 10,000 mm^3), deep cerebellar nuclei volume was approximately 50% higher among eutherians than among metatherians and monotremes of the same cerebellar cortical volume. Within metatherians (Fig. 10b), scaling of deep cerebellar nuclei with cerebellar cortical volume appeared to be similar for diprotodontia and other metatherians.

4. Discussion

4.1. Limitations of the present study

The present study is based mainly on analysis of sectioned brains in either the Nelson Brain Collection, held at the Australian National Wildlife Collection in Canberra, or photomicrographs of sectioned brains that are part of the Neuroscience Library Collection online. All of the sectioned brains from those two collections have been fully dehydrated and embedded in either paraffin or celloidin. Some data has been drawn from analysis of undehydrated frozen sections (one echidna, one platypus and the tammar wallaby in the author's collection). All measurements were corrected back to pre-dehydration dimensions for all species. This was done either by calculation of a correction factor from comparison of known brain volume with calculated volume after dehydration, or by application of an average shrinkage factor when pre-dehydration brain size was unknown. The average shrinkage factor (0.738 in a linear dimension; 0.545 in area; and 0.402 in volume) was calculated from 47 mammalian species where pre- and post-embedding brain volume was known. The shrinkage factor does not correlate significantly with brain size (r^2 = 0.012; df = 45; p = 0.228).

Calculation of the volume of functionally significant regions (e.g. flocculonodular lobe) is necessarily based on anatomical criteria, rather than functional analysis. In the case of the flocculus and nodule (10Cb), the demarcation with adjacent regions requires identification of the landmark posterolateral fissure. It is possible that the boundary between vestibulocerebellar cortex and other functional divisions may not always lie at the posterolateral fissure, but clearly it is not possible to make functional assessment of vestibulocerebellar boundaries in archived museum material. For this reason one should be cautious about interpreting findings of low flocculonodular lobe volume in some

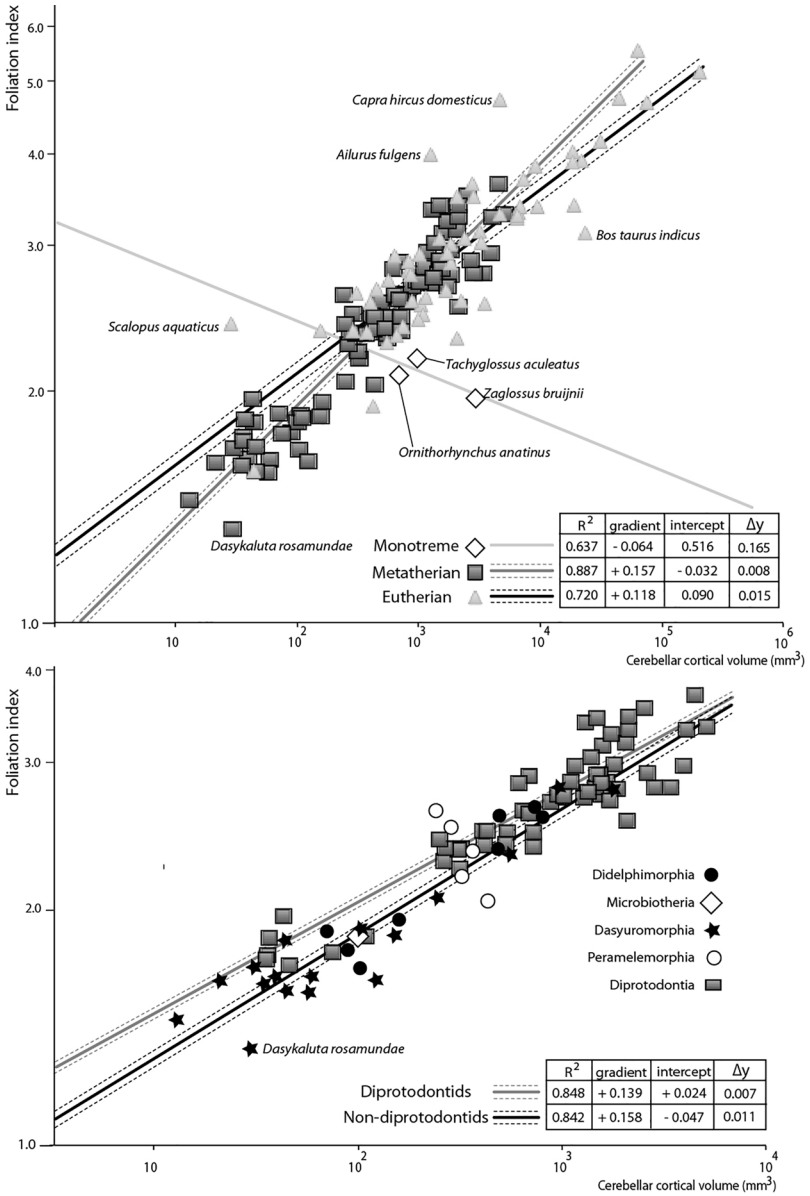


Fig. 6. Plots of foliation indices (fi; PSC/ESC) against total cerebellar cortical volume in mm^3 for monotremes, metatherians, and eutherians (a) and among metatherians (b). The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Among therians, foliation increases with cerebellar cortical volume, with some significant outliers as labelled. The three monotreme species have poorly foliated corpus cerebelli for their cerebellar cortical volume, particularly the long-beaked echidna *Zaglossus bruijnii*. Among metatherians (b), diprotodontia have higher foliation indices than didelphimorphia and dasyuromorphia of similar cerebellar cortical volume. Note that the intercepts in the table in a) are $\log_{10}$ of the foliation index for a cerebellar cortical volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regression) is also $\log_{10}$ of the foliation index.

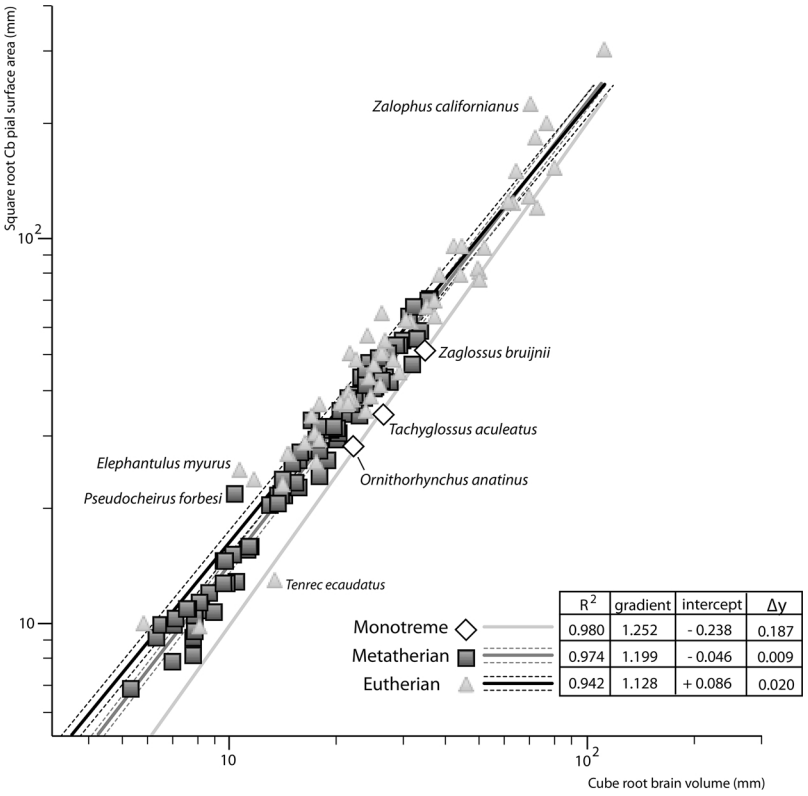
species (e.g. *Zaglossus bruijnii*). Similarly, delineation of the boundaries of the pontine and deep cerebellar nuclei of necessity is based on anatomical landmarks. Delineating boundaries for these structures requires distinguishing a transition from a region rich in large or medium-sized neuronal cell bodies to adjacent white matter by eye in Nissl-stained sections.

The variables analysed in the present study were those that could be extracted from measurements of archived dehydrated, sectioned and stained tissue. Clearly it would be preferable to analyse neuronal and non-neuronal number in these brain regions using isotropic fractionation (Herculano-Houzel, 2010; Dos Santos et al., 2017) or similar, but that is not possible with the available material and NeuN does not label Purkinje cells (Herculano-Houzel, 2010). Nevertheless, the results provide important insights into scaling of anatomical parts of the cerebellum in different mammals.

4.2. Structural and functional differences in the brains of prototherians and therians

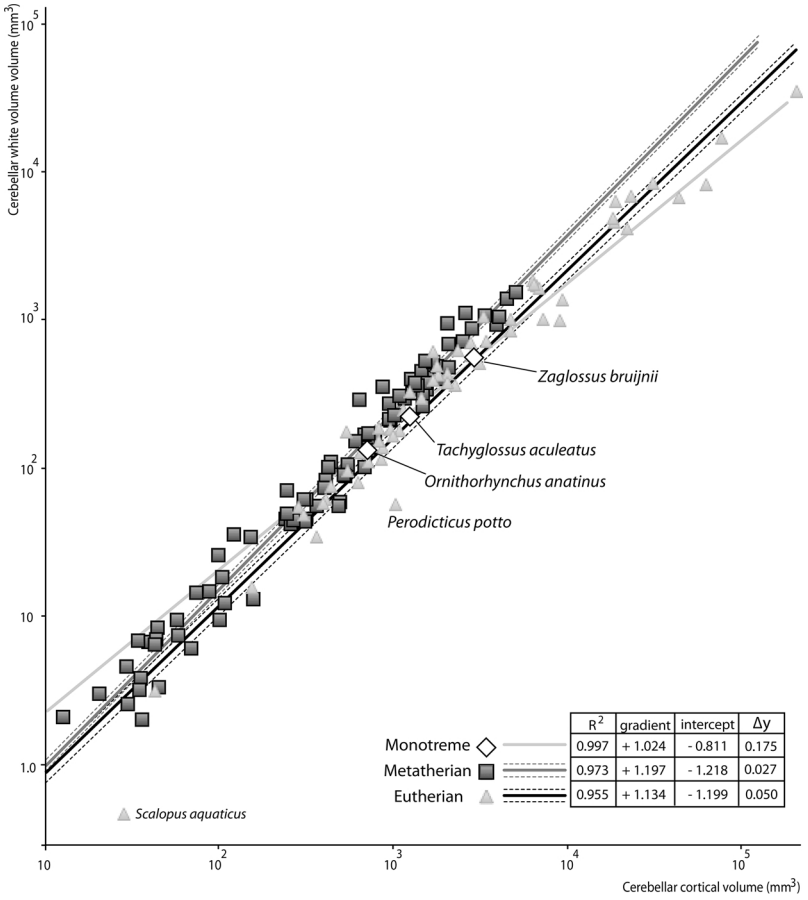
The prototherian and therian lineages diverged at least by the early Jurassic (160 to 220 million years ago; Musser, 2003; Phillips et al., 2009; Yu et al., 2012). This prolonged period of divergent brain evolution is reflected in structural brain differences between monotremes and therians. A previous quantitative study of the forebrain structure across mammals showed that the proportion of subiso- and periallocortical white matter in monotremes is substantially lower than that in the forebrains of therians (Ashwell and Gurovich, 2019). This suggests that monotremes have fewer associational, commissural and/or projection connections in their forebrains compared to similarly sized forebrains of therian mammals.

Findings in the present study indicate that monotremes have smaller corpus cerebelli for their brain size, lower cerebellar foliation and smaller cerebellar cortical surface area compared to therians. On the other hand, the volume of deep white matter in the cerebellum of monotremes is comparable to that in therians with a similar cerebellar cortical volume, and scaling of pontine nuclear volume to cerebellar cortical volume is identical in monotremes and therians. Deep cerebellar nuclei are comparable in volume in the monotremes to metatherians of a similar cerebellar cortical volume, but smaller than eutherian mammals.



	R^2	gradient	intercept	Δy
Monotreme	0.980	1.252	- 0.238	0.187
Metatherian	0.974	1.199	- 0.046	0.009
Eutherian	0.942	1.128	+ 0.086	0.020

Fig. 7. Plot of the square root of cerebellar pial surface area (√PSC) against the cube root of brain volume in mm for monotremes, metatherians, and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Outlier species have been labelled (*Elephantulus myurus*, *Pseudocheirus forbesi*, *Tenrec ecaudatus*, *Zalophus californianus*). The three species of monotreme (labelled) have low cerebellar PSC for their brain size, compared to therians. Note that the intercepts in the included table are $\log_{10}$ of the square root of PSC for a cube root of brain volume value of 1 mm. The value for Δy (indicating 95% confidence interval for the regression) is also $\log_{10}$ of the square root of PSC.



	R^2	gradient	intercept	Δy
Monotreme	0.997	+ 1.024	- 0.811	0.175
Metatherian	0.973	+ 1.197	- 1.218	0.027
Eutherian	0.955	+ 1.134	- 1.199	0.050

Fig. 8. Plot of the cerebellar white matter volume against cerebellar cortical volume for monotremes, metatherians, and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Cerebellar white matter volumes for the three monotreme species are similar to therians of the same cerebellar cortical volume. For therian species, cerebellar white matter volume scales with positive allometry (gradients of 1.197 and 1.134 for metatherians and eutherians), but close to isometry for the monotremes (gradient of 1.024). Two outlier eutherian species with low cerebellar white matter volume for their cerebellar cortical volume are labelled (*Scalopus aquaticus* and *Perodicticus potto*). Note that the intercepts in the included table are $\log_{10}$ of the cerebellar white matter volume for a cerebellar cortex volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regressions) is also $\log_{10}$ of the deep cerebellar white matter volume in mm^3.

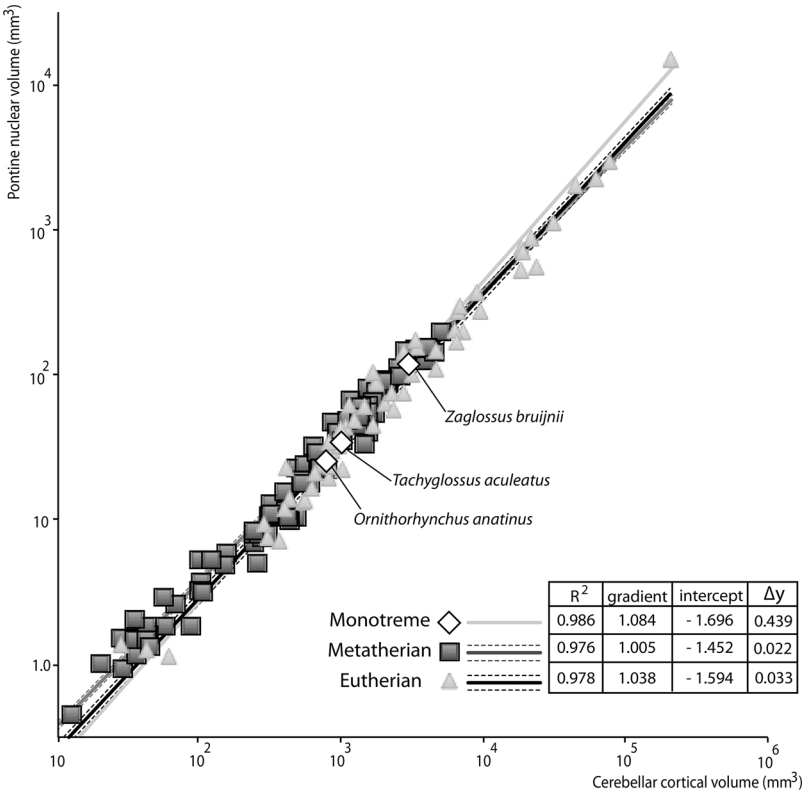


Fig. 9. Plot of the pontine nuclear volume (actually basilar pontine and pontine reticulotegmental nuclei) against cerebellar cortical volume for monotremes, metatherians, and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Pontine nuclear volumes for 3 monotreme species are similar to therians of the same cerebellar cortical volume. For therian species, pontine nuclear volume scales close to isometry (gradients of 1.005 and 1.038 for metatherians and eutherians), but slightly above isometry for the monotremes (i.e. positive allometry; gradient of 1.084). Note that the intercepts in the included table are $\log_{10}$ of the pontine nuclear volume for a cerebellar cortex volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regressions) is also $\log_{10}$ of the pontine nuclear volume in mm^3.

The cerebellar white matter findings suggest that the monotreme cerebellum has comparable internal connectivity to the therians, despite being smaller for a given brain size. This stands in contrast to our previous finding that the volume of white matter beneath the iso- and periallocortical grey matter was reduced in monotremes compared to therians of similar cortical grey matter volume.

The findings also suggest that the long-beaked echidna has expanded cerebellar hemispheres compared to its nearest relative, the short-beaked echidna, and the platypus. This is difficult to explain on behavioural grounds, because both the long- and short-beaked echidnas have similar metabolism and occupy similar insectivorous niches (Dawson et al., 1978; Opiang, 2009) and there is no obvious significant behavioural distinction between short and long-beaked forms.

4.3. Structure and function of the cerebellum in metatherians versus eutherians

It is now recognised that extant metatherians cannot be regarded as primitive and should not be seen as extant proxies for ancestral mammals (Dos Santos et al., 2017). With respect to cerebellar morphology, metatherians and eutherians have much in common. Total cerebellar volume, cerebellar hemispheric volume and cerebellar surface area scale almost identically amongst both metatherians and eutherians, and foliation increases with cerebellar size as much among metatherians as eutherians, although metatherians never reach the cerebellar volumes and degree of foliation seen among eutherians.

However, there are some differences between metatherians and eutherian mammals in cerebellar morphology. Dos Santos et al. (2017) noted that the cerebella of metatherians have more neurons than non-primate eutherian cerebella of a similar mass. In the present study, it was found that metatherian cerebella have slightly larger flocculonodular lobes compared to eutherian cerebella of a similar volume, cerebellar white matter volume is slightly higher for a given volume of cerebellar cortex among metatherians than eutherian mammals, and deep cerebellar nuclear volume is slightly lower among metatherians than among eutherian mammals for a given volume of cerebellar cortex. Taken together, these findings suggest that there may be subtle but functionally important differences between metatherians and eutherian mammals in the proportion of cerebellum devoted to vestibulocerebellar function and in the way that the cortex and deep cerebellar nuclei are connected.

Related to this is the observation that proportional hemispheric expansion [i.e. the proportion of cerebro(ponto)cerebellum of the entire cerebellum] is very low even among metatherian species with large cerebella (e.g. cerebellar volume of 10^4 mm^3, *Macropus rufus*), whereas hemispheric expansion is better developed among some smaller metatherian species with cerebellar volumes of less than 10^3 mm^3 (e.g. *Acrobates pygmaeus*, *Gymnobelideus leadbeateri*, *Petaurus australis*, *Tarsipes rostratus*). Cerebellar hemispheric [i.e. functionally cerebro(ponto)cerebellar] expansion is a prominent feature among particular groups of eutherian mammals (e.g. primates, pinnipeds, cetaceans) and may have arisen independently multiple times (Smaers et al., 2018). Enlarged cerebellar hemispheres among eutherian mammals have been linked with a capacity for vocal production learning, and expansion of cortico(ponto)cerebellar circuits involved in visual grasping and reaching, and echolocation (Smaers et al., 2018).

4.4. Structure and function of the cerebellum in diprotodontid versus non-diprotodontid metatherians

In terms of species numbers, modern metatherians have about one-tenth the diversity of modern eutherians, but metatherians encompass almost the same range of lifestyles as seen in eutherians, with the exception of bats and cetaceans (Archer and Kirsch, 2006). The Diprotodontia is the most diverse metatherian order, as determined by number of species. The order includes ten living families and over 115 living species. This group underwent an adaptive radiation in the Miocene that allowed them to occupy a strikingly diverse range of niches from arboreal folivores and frugivores (possums, tree kangaroos, the koala), through nectar and pollen-feeders (honey possum and feathertail glider), terrestrial grazers (kangaroos), fungi-eaters (bettongs), to burrowing root and tuber-eaters (wombats). Modes of locomotion and

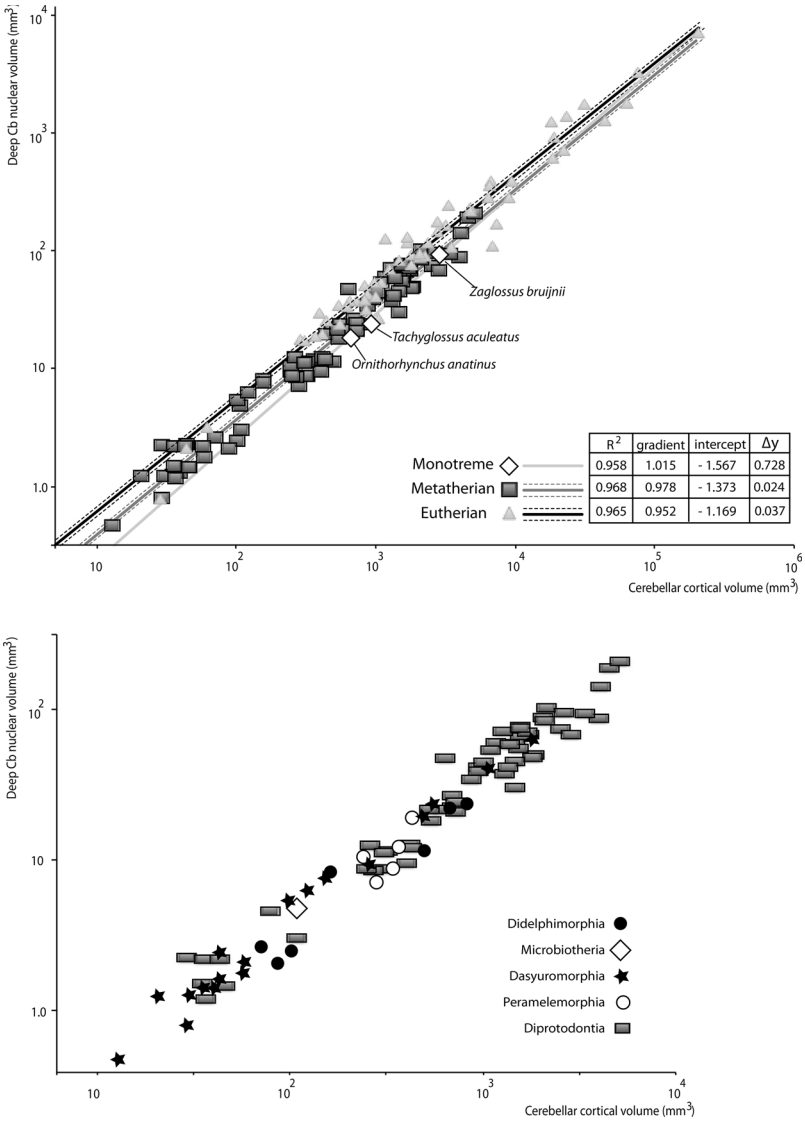


Fig. 10. Plot of the volume of the deep cerebellar nuclei against cerebellar cortical volume for monotremes, metatherians, and eutherians. The 95% confidence intervals for the linear regressions are shown as dashed lines for eutherian and metatherian data. Note that the intercepts in the included table are $\log_{10}$ of deep cerebellar nuclei volume in mm^3 for a cerebellar cortical volume of 1 mm^3. The value for Δy (indicating 95% confidence interval for the regressions) is also $\log_{10}$ of the deep cerebellar nuclear volume in mm^3. Deep cerebellar nuclei volumes for 3 monotreme species are similar to metatherians of the same cerebellar cortical volume, but both monotremes and metatherians have lower deep cerebellar nuclei volume than eutherians of a similar cerebellar cortical volume. For all three mammalian groups, deep cerebellar nuclei volume scales close to isometry (gradients of 1.015, 0.978 and 0.952 for monotremes, metatherians and eutherians).

habitat types in the group are just as diverse as their food sources, reflecting the enormous success of the diprotondontid lineage. Some diprotodontids (e.g. phalangerids) also show convergent evolution with some eutherians (e.g. rodents) in the area of trigeminal pathway specialization, with a barrel-field like region of the primary somatosensory cortex (Weller, 1972, 1993). Other diprotodontids have been shown to have specialization of cortical fields representing other body parts (e.g. the behaviourally important digit 4 of the manus of *Dactylopsila trivirgata*; Huffman et al., 1999). Diprotodontids are also more encephalized and neurologically specialised than other metatheria.

Some of the present findings indicate structural and possibly functional cerebellar differences between diprotodontids and other metatherians. In particular, there appeared to be higher hemispheric specialisation and greater foliation among diprotodontids compared to non-diprotodontids, particularly at smaller cerebellar volumes. Those diprotodontid metatherians exhibiting the greatest hemispheric enlargement were small nectar and pollen-feeding possums, one with unusual auditory specialisations (e.g. *Acrobates pygmaeus*; Aitkin and Nelson, 1989) and smaller petauroid gliders (*Petaurus australis*, *Petaurus norfolcensis*; Goldingay and Quin, 2001; Goldingay and Taylor, 2009). At present there are no data quantifying manual dexterity across a range of Australian metatherians, but it is possible to identify special locomotor and manual attributes among those metatherian species, that may be associated with their cerebellar hemispheric enlargement.

The small metatherian feathertail glider (*Acrobates pygmaeus*) feeds on fruit, manna, honeydew, pollen and small arthropods (Goldingay and Kavanagh, 1995). Apart from its adept gliding (Pridmore and Hoffmann, 2014), the feathertail glider uses diagonal-sequence couplets gait and hallucal grasping, and shows high maternal investment in its young long after weaning, which are all features in convergence with primates (Karantanis et al., 2015; Youlatos et al., 2018). In fact, although taxonomically distinct, some authors have used the feathertail glider as a model of early primate locomotion (Karantanis et al., 2015). The tiny, arboreal honey possum (*Tarsipes rostratus*) is also able to climb with the assistance of the prehensile tail and an opposable hallux on its long hindfoot that is functionally similar to the pes of arboreal primates. Small gliders of genus *Petaurus*, not only navigate a complex three-dimensional environment, but also have primate-like quadrupedalism and complex social structure (Goldingay and Quin, 2001; Shapiro and Young, 2010). Arboreality and gliding require high levels of dynamic and postural control of musculature (Byrnes and Spence, 2011) and this may explain the high degree of cerebellar hemispheric specialisation seen in *Acrobates*, *Tarsipes* and *Petaurus*.

It is possible that the expansion of the cerebellar hemispheres in small, agile diprotodontids occurred in concert with invasion of new niches, but the paleoneurological fossil record is too patchy at present to allow testing of this hypothesis.

5. Concluding remarks

Although the cerebellum presents a stereotypical structure in all mammals, quantitative analysis indicates that there are structural differences in cerebellar morphology between monotremes and therians, between metatherians and eutherian mammals, and between diprotodontid and non-diprotodontid metatherians. The observed differences between diprotodontid and non-diprotodontid metatherians may reflect specialisation of cerebellar systems for arboreality, prehensility and gliding.

Acknowledgements

I am extremely grateful to Emeritus Professor John Nelson of Monash University and Dr Leo Joseph of CSIRO, who kindly gave permission to photograph and analyse the sectioned and stained monotreme and metatherian brains from the Nelson Brain Collection at the Australian National Wildlife Collection in Canberra. The study would also not have been possible without the excellent online resources of the neurosciencelibrary.org. I would like to thank Damien McKay for his assistance with the analysis. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.zool.2020.125753.

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